How to Interpret Tinnitus Functional Index Scores: A Proposal for a Grading System Based on a Large Sample of Tinnitus Patients

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Objectives: The Tinnitus Functional Index (TFI) is considered the gold standard in measuring tinnitus severity. The aim of the study was to establish reference values to improve the interpretability of TFI scores.

Design: Results from 1114 patients with tinnitus were retrospectively analyzed. The participants were consecutive patients who attended our tertiary referral Ear, Nose, and Throat Center. The eligibility criteria were: at least 18 years old, persistent tinnitus, completed pure-tone audiometry, and answered all 25 items on the TFI. Hearing status (normal hearing vs. hearing impairment) was established according to the recommendation of the Bureau International d'Audiophonologie. Means (M) and SD on the TFI were the basis for grading tinnitus severity on four levels: low, lower moderate, upper moderate, and high. To gauge individual scores in clinical practice, percentiles are also proposed.

Results: All 1114 patients (586 women and 528 men) were Caucasian and aged from 19 to 87 years (M = 50.96; SD = 13.10 years). Tinnitus duration ranged from 0.5 to 50 years (M = 7.17; SD = 7.71 years). There were 258 patients with normal hearing and 856 patients with hearing loss. A score of above 65 points on TFI was established as the cutoff point for diagnosing high tinnitus severity. A regression model associating tinnitus severity with gender, age, tinnitus duration, and hearing loss was statistically significant: $F(4,1109) = 8.99$; $p < 0.001$, but the effect was very small ($R^2_{adj} = 0.028$) and only gender and age were associated with TFI global score, while tinnitus severity was not related to tinnitus duration or hearing loss.

Conclusions: The reference values proposed here support those reported previously by Meikle et al. They are empirically based and can be used as benchmarks in clinical practice and scientific research. They make it possible to assess tinnitus severity, evaluate individual scores, and categorize individuals with tinnitus. This allows researchers to set inclusion or exclusion criteria when assigning patients to specific groups during clinical trials involving tinnitus intervention strategies.

Key words: Interpretability, Norming, Normative values, Reference values; Tinnitus, Tinnitus functional index.

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INTRODUCTION

Subjective tinnitus is the most common form of tinnitus (Tunkel et al. 2014), defined as a phantom sound sensation without an identifiable sound source (Moller 2011b; Jastreboff 2015). Tinnitus reduces quality of life and imposes restrictions on everyday functioning (Zeman et al. 2014). Epidemiological studies reveal that the prevalence of tinnitus varies from 4.4 to 15.1% in adults (Moller 2011a), although some studies have shown it can reach 11.9 to 30.3% (McCormack et al. 2016). Tinnitus affects not only adults but also children, and among them the prevalence varies from 6 to 34% (Savastano 2007; Skarżyński et al. 2011; Piotrowska et al. 2015; Skarżyński et al. 2016).

An objective measure for quantifying tinnitus does not exist. Because of its subjective nature, measurement of tinnitus poses many challenges and its diagnosis and evaluation depend on subjective methods such as visual analog or numeric rating scales (Raj-Koziak et al. 2018) and questionnaires (Fackrell et al. 2014). Such methods can be categorized as patient-reported outcome measures (PROMs). The use of PROMs is increasing, associated with patient involvement in the organization and delivery of healthcare (Black et al. 2014). A PROM is a measure of a certain aspect of a patient's health status that comes directly from the patient, as defined by the US Department of Health and Human Services FDA Center for Drug Evaluation and Research (2006). PROMs can be general or disease-specific, but need to be both validated and psychometrically robust (Aaronson et al. 2002; Terwee et al. 2007; Mokkink et al. 2010).

There are many self-report measures used in clinical practice and in research assessing tinnitus, and multi-item questionnaires play a key role. Hall et al. (2016) showed that there were at least 29 different tinnitus-related questionnaires used in clinical practice. However, all have particular shortcomings, and therefore, new tools are still being created (Tyler et al. 2014; Skarżyński et al. 2018). Some of the questionnaires commonly applied by clinicians have been recommended for diagnostic use, for example, the Tinnitus Handicap Inventory, Tinnitus Handicap Questionnaire, and Tinnitus Questionnaire (Landgrebe et al. 2012; Skarzynski et al. 2017). Although there is much evidence on the reliability and validity of these tools, evidence on their responsiveness is much more limited. Responsiveness is a measure of the ability to detect a clinically relevant change over time, which is especially important in clinical trials when assessing the effectiveness of an intervention for tinnitus (Kamalski et al. 2010; Londero & Hall 2017).

The Tinnitus Functional Index (TFI), developed by Meikle et al. (2012), stands out among other tinnitus-related questionnaires because of its responsiveness. The considerable advantage of TFI is its wide coverage of many aspects of tinnitus

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and its ability to measure tinnitus severity in multiple domains: sleep, emotions, cognition, sense of control, quality of life, and others. The overall score is an aggregate measure of all the negative impacts of tinnitus on various aspects of life. The tool provides reliable and valid measures of tinnitus severity, it has a clear factor structure, and has been adapted to many languages. Unlike other tinnitus-related questionnaires, there is evidence for sensitivity of the TFI to intervention-related change (Meikle et al. 2012; Skarżyński et al. 2018).

However, as with other tinnitus-related questionnaires, limited information is available on the *interpretability* of TFI scores. Interpretability is the degree to which one can assign a qualitative meaning to a quantitative score (Aaronson et al. 2002). It is important for both practitioners and researchers to be able to reliably grade tinnitus severity, give a clinical meaning to the numerical score, and, based on this, assign patients with tinnitus to a specific category. In other words, a grading system of tinnitus severity is needed that allows TFI scores to be clinically interpreted.

The authors of the TFI did provide a preliminary guide for interpreting results. Scoring was based on grading tinnitus severity by the patients themselves, who were asked "How much of a problem is your tinnitus?" When tinnitus was not a problem the TFI global score was on average M = 13.6 points; if a small problem M = 20.7; if a moderate problem M = 42.4; if a big problem M = 64.0; and if very big problem M = 77.6. These results supported a three-category grading system: mild tinnitus (score < 25 points); significant problems with tinnitus (25 to 50 points); and severe tinnitus (>50 points), but this grading was labeled "preliminary" because it was based on a limited sample of subjects (groups ranged from 8 to 113 patients, with 347 patients in total). Henry et al. (2016) stressed that because TFI is utilized in numerous clinical trials, relevant and reliable guidelines should be provided for administering the tool and interpreting its results, and they referred to the grading system proposed by Meikle et al. (2012).

To allow TFI scores to be clinically interpreted, grading of tinnitus severity requires the setting of strictly defined values. Such values are called "normative" (or "the norm"), but in a medical context "normal" is strongly associated with "healthy" or "nonpathological," whereas this concept does not sit comfortably with individuals with tinnitus (who cannot be said to be completely healthy). In statistics, "normal" is interpreted as being typical, or most commonly encountered, and is defined by an interval: that is, threshold values between which a specified percentage (usually 95%) of individuals fall. To avoid possible confusion, use of the phrase "reference values" seems more appropriate than "normative values." The term reference values was coined by Gräsbeck and Saris (1969), referring to establishment and use of relevant data for interpreting medical observations. Reference values can be derived from healthy individuals or, as Boyd (2010) suggested, such values can be established for a given reference population. Gräsbeck (2004) emphasized that in such a case the criteria for the selection of the individuals needs to be given. In the case of tinnitus severity, the only reasonable reference group is a large group of patients experiencing tinnitus.

In their review of tools for quantitative assessment of tinnitus, Langguth et al. (2011) stressed the importance of establishing normative values. They said: *Norming means obtaining information about the distribution of measures in a target population in the form of means, standard deviations, or percentiles. Normative data allow placement of the score of an individual in context of a target population*. Evidently, they understood norming as facilitating interpretability. Their article describes two approaches that can be used to determine reference values for TFI scores: reference values can either be given in terms of the number of standard deviations away from the population mean, or the results can be reported in terms of percentiles. In this work, we use both options.

The present study aims to provide guidelines for reliably grading tinnitus severity as measured with TFI. The guidelines are supported by strong empirical evidence derived from a large sample of patients with tinnitus.

MATERIALS AND METHODS

Subjects

Our subjects were from an original database containing medical data from 1247 adult patients with tinnitus, consecutively admitted to our tertiary referral center due to tinnitus and/or hearing problems, who agreed to fill in a tinnitus questionnaire as part of their examination. Our retrospective data were screened for compliance with the following criteria: existence of persistent tinnitus (duration of at least 6 months); documented hearing threshold test based on clinical pure-tone audiometry; and have completed the Tinnitus Functional Index questionnaire, answering all 25 items. Figure 1 is the flowchart for patient screening, showing how the final size of the study group was established.

Measures

Tinnitus Functional Index • The TFI is a self-report questionnaire measuring severity and negative impacts of tinnitus (Meikle et al. 2012). It allows the clinician to comprehensively assess how severe tinnitus is in multiple domains. The TFI

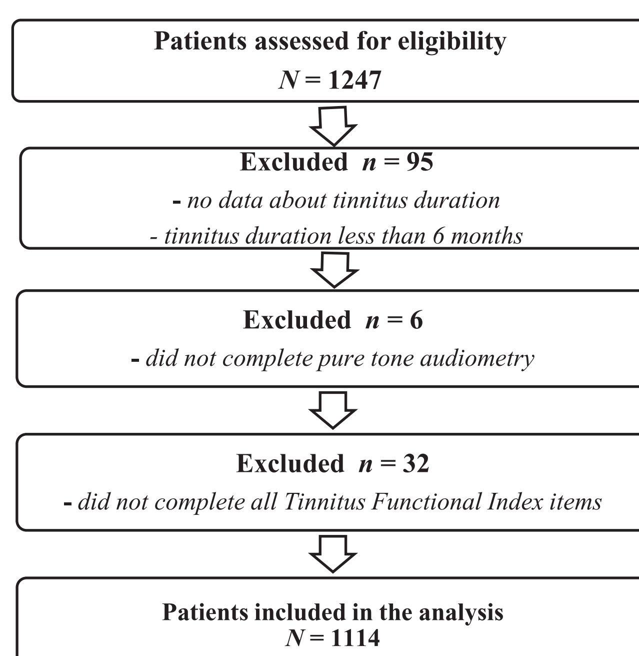

Fig. 1. Flowchart of patient screening.

includes eight subscales: intrusiveness, sense of control, cognition, sleep, auditory, relaxation, quality of life, and emotional. The subscale Quality of Life consists of four items; other subscales consist of three items, giving 25 items in total. The global score, as with scores in each subscale, can range from 0 to 100 points. The higher the score, the higher the severity of tinnitus and the greater the negative impact on everyday life. The Polish version of the TFI questionnaire was licensed from the Oregon Health & Science University for use in clinical and scientific research (from April 2016 to April 2021). The psychometric properties of the Polish TFI were comparable to the original version (internal consistency was very high with a Cronbach's α of 0.96; factor structure was confirmed; convergent validity was supported by a strong correlation with the Tinnitus Handicap Inventory of 0.78) as documented by Wrzosek et al. (2016).

Pure-tone Audiometry • All measurements were conducted in a soundproof booth by an experienced technician using a Madsen Itera II diagnostic audiometer (GN Otometrics, Denmark) with calibrated TDH-39P earphones (Telephonics, New York, USA). Hearing thresholds were determined for air conduction in the right and left ears at frequencies of 125, 250, 500, 1000, 2000, 4000, and 8000 Hz and for bone conduction at frequencies of 250, 500, 1000, 2000, and 4000 Hz. Pure-tone average was calculated as the average of hearing threshold levels at 500, 1000, 2000, and 4000 Hz according to the Bureau International d'Audiophonologie (BIAP) recommendation (1996). Hearing loss according to BIAP is defined as an average pure tone loss of more than 20 dB at 500, 1000, 2000, and 4000 Hz (the worse ear being considered). Degrees of hearing loss were assessed as mild (21 to 40 dB), moderate (41 to 70 dB), severe (71 to 90 dB), or very severe (91 to 120 dB) again according to BIAP standards.

Ethical Consideration

The protocol of this retrospective study was approved by the local Bioethics Committee (approval no. KB.IFPS:17/2018).

Statistical Analysis

Descriptive statistics (M, mean; SD) were calculated for the TFI scores and the distribution of TFI global scores was displayed. A spacing was established of –1 SD below the mean and +1 SD above, and on this basis, four categories of scores indicating various levels of tinnitus severity were specified: scores below M – 1SD were called "low"; scores between M – 1SD and M "lower moderate"; scores between M and M + 1SD "upper moderate"; and above M + 1SD "high."

Percentiles were also calculated in the form of a table showing cumulative percentage of the TFI global score (Kline 2000; Urbina 2004).

Linear regression modeling was used to investigate whether there was a statistically significant association between tinnitus severity and gender, age, tinnitus duration, and hearing loss. Zero-order, part, and partial correlations were examined to investigate the individual impact of each predictor on tinnitus severity. Effect size for significant predictors was calculated and interpreted as small (r = 0.02), medium (r = 0.15), or large (r = 0.35) (Cohen 1988, 1992). Diagnostics of regression residuals were additionally conducted. Statistical analyses were performed using SPSS v. 24 (IBM SPSS, IBM Corp, 2016, Armonk, New York, USA).

RESULTS

There were 1114 patients: 586 women and 528 men, aged from 19 to 87 years (M = 50.96; SD = 13.10). The mean age for women was 50.83 years (SD = 13.05) and for men it was similar (M = 51.10; SD = 13.16). Tinnitus duration ranged from 0.5 to 50 years, with mean duration 7.17 years (SD = 7.71); there was little difference between women (M = 7.08; SD = 7.74) and men (M = 7.27; SD = 7.58).

The pure-tone average for the right ear was 38.78 dB HL (SD = 29.93) and for the left ear 37.98 dB HL (SD = 28.71) for air conduction; for bone conduction, the comparable figures were 26.70 dB HL (SD = 21.42) and 27.06 dB HL (SD = 21.08). There were 258 patients (123 women and 135 men) with normal hearing and 856 patients (463 women and 393 men) with hearing loss. In patients with normal hearing, the mean air conduction threshold was 10.80 dB HL (SD = 5.24) for the right ear and 11.37 dB HL (SD = 5.11) for the left; the mean bone conduction thresholds were 7.41 dB HL (SD = 5.22) for the right ear and 8.13 dB HL (SD = 5.43) for the left. In patients with hearing loss, the mean air conduction threshold was 47.12 dB HL (SD = 29.17) for the right ear and 46.00 dB HL (SD = 28.06) for the left; the mean bone conduction thresholds were 32.52 dB HL (SD = 21.04) for the right ear and 32.77 dB HL (SD = 20.71) for the left.

Descriptive statistics for the global TFI score were calculated from statistical analysis of the data (Table 1).

The patients achieved the highest scores on the subscale of Intrusiveness; the lowest on the Cognition subscale. The mean global TFI score was 41.80 points (SD = 23.42) and it was close to the median of 40.40 points; the SD was more than 50% of the mean score. The distribution of the TFI global score was assessed in terms of normality. A Kolmogorov–Smirnov test was statistically significant (K-S = 0.06; $p < 0.001$). Skewness was 0.30 (SE = 0.07), and kurtosis was –0.70 (SE = 0.15). The distribution of the scores was slightly positively skewed, which means there were more low scores on the left side of the histogram. This indicates that a large fraction of patients with tinnitus experience lower than average severity, while the number of subjects with high tinnitus severity is smaller. The distribution was platykurtic, tending to be flatter than normal.

Figure 2 is the distribution of the TFI global scores.

Due to the fact that the distribution of the TFI global score did not quite conform to a normal curve, an attempt was made to improve the shape of the distribution. In the case of positive skewness, two transformations are recommended: square root and log transformation (Tabachnick & Fidell 2007). However, neither of these gave satisfactory results: they did not make the skewness or kurtosis values come close to zero and did not improve the shape of the distribution (the square root transformation slightly improved the kurtosis but worsened the skewness, while log transformation worsened both). We decided to leave the distribution in its original form.

The determination of reference values was done on the basis of M and SD. For all subjects the reference values were calculated in the following way:

1. For low scores, the interval between 0 and M – 1SD was chosen. The upper limit was calculated as 41.80 – 23.42 = 18.38 and rounded to 18. Scores between 0 and 18 points can be considered low.

TABLE 1. Descriptive statistics for TFI scores (N = 1114)

	Min	Max	M	SD	Q1	Me	Q3	Skewness	Kurtosis
Intrusiveness	0	100	57.97	25.05	40.00	56.67	76.67	−0.14	−0.88
Sense of control	0	100	40.36	29.31	16.67	36.67	60.00	0.41	−0.85
Cognition	0	100	34.64	27.92	10.00	30.00	56.67	0.53	−0.69
Sleep	0	100	38.60	32.60	10.00	33.33	66.67	0.44	−1.12
Auditory	0	100	42.87	28.76	16.67	43.33	66.67	0.15	−1.04
Relaxation	0	100	44.97	29.38	20.00	43.33	67.50	0.18	−1.01
Quality of life	0	100	38.17	27.67	12.50	37.50	60.00	0.32	−0.89
Emotional	0	100	38.03	29.24	13.33	33.33	60.00	0.49	−0.83
TFI global score	0	100	41.80	23.42	22.80	40.40	58.80	0.30	−0.70

M, mean; Min, minimum, Max, maximum; Me, median; Q1, lower quartile; Q3, upper quartile; TFI, Tinnitus Functional Index.

2. For lower moderate scores, the interval between M – 1SD to M was chosen, so scores between 18 and 42 points can be considered lower moderate.
3. For upper moderate scores, the interval between M to M + 1SD was chosen, so scores between 42 and 65 points can be considered upper moderate.
4. For high scores, the interval between M + 1SD and 100 was chosen, so scores higher than 65 points can be considered high.

An equivalent meaning for the TFI scores is provided by percentiles. These are shown in Table 2.

An example interpretation of the percentile result is as follows: a subject who achieved a global TFI score of 67 points scores in the 85th percentile. In this context, a result of 67 points can be interpreted as high.

Several characteristics can potentially be associated with tinnitus severity (gender, age, tinnitus duration, and hearing loss) and these were investigated using multiple linear regression (Table 3).

The regression model was statistically significant: $F(4,1109) = 8.99$; $p < 0.001$, but only 2.8% of the total variance of the TFI global score ($R^2_{adj} = 0.028$) was accounted for by predictors included in the model. The results show that gender ($p = 0.002$) and age ($p < 0.001$) are significantly associated with TFI global score, while tinnitus duration and hearing loss do not influence tinnitus severity. The distribution of the residuals in the model was inspected by a normal probability plot. All standardized regression residuals were in the range ±3 SD. The Durbin–Watson statistic was 1.82, showing there was no autocorrelation in the residuals.

Despite the regression analysis showing statistically significant effects of gender and age, the mean difference for gender lacked clinical significance. Women (M = 43.79; SD = 24.14) scored slightly higher than men (M = 39.59; SD = 22.40), but the mean difference was only about four points and the effect size was 0.081. The effect size for age was even smaller (0.024) and again statistical significance did not translate into clinical significance.

We found that the effect of hearing loss on tinnitus severity was not significant and nearly zero. Patients with normal hearing reported slightly lower tinnitus severity (M = 40.37; SD = 22.89) than patients with hearing loss (M = 42.23; SD = 23.57). The

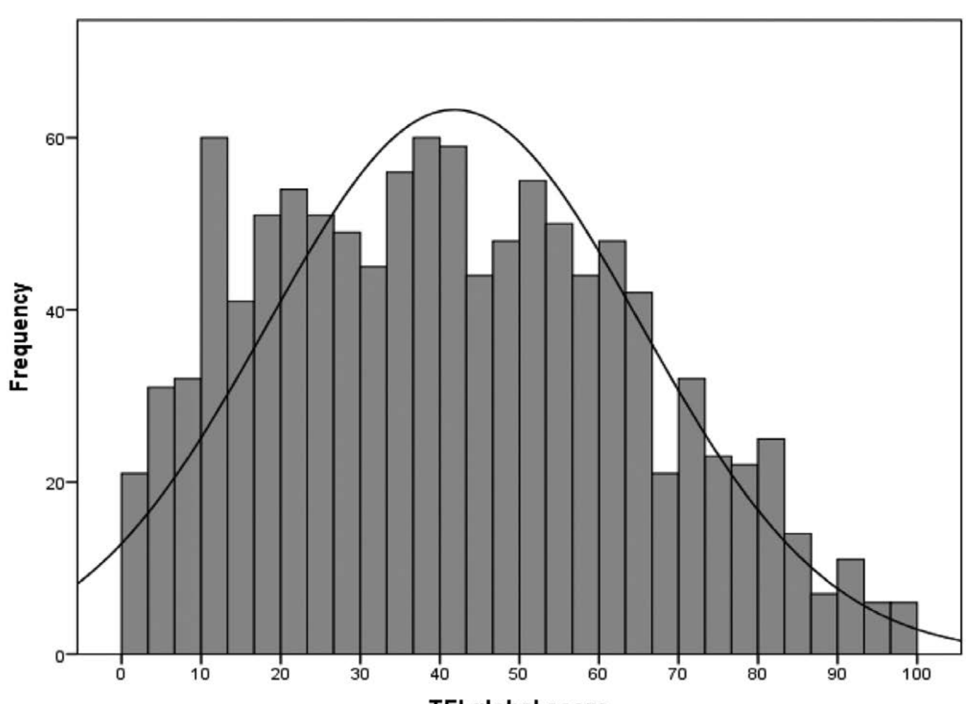

Fig. 2. Distribution of TFI global scores. TFI, Tinnitus Functional Index.

TABLE 2. Frequency distribution for global TFI score (N = 1114)

TFI global score	All patients Frequency	Percent	Cumulative frequency	Cumulative percent
0	4	0.36	4	0.36
1	3	0.27	7	0.63
2	8	0.72	15	1.35
3	6	0.54	21	1.89
4	9	0.81	30	2.69
5	9	0.81	39	3.50
6	13	1.17	52	4.67
7	9	0.81	61	5.48
8	14	1.26	75	6.73
9	6	0.54	81	7.27
10	11	0.99	92	8.26
11	20	1.80	112	10.05
12	19	1.71	131	11.76
13	13	1.17	144	12.93
14	18	1.62	162	14.54
15	10	0.90	172	15.44
16	13	1.17	185	16.61
17	6	0.54	191	17.15
18	24	2.15	215	19.30
19	16	1.44	231	20.74
20	19	1.71	250	22.44
21	8	0.72	258	23.16
22	16	1.44	274	24.60
23	16	1.44	290	26.03
24	22	1.97	312	28.01
25	11	0.99	323	28.99
26	18	1.62	341	30.61
27	17	1.53	358	32.14
28	18	1.62	376	33.75
29	8	0.72	384	34.47
30	20	1.80	404	36.27
31	9	0.81	413	37.07
32	15	1.35	428	38.42
33	7	0.63	435	39.05
34	17	1.53	452	40.57
35	15	1.35	467	41.92
36	24	2.15	491	44.08
37	17	1.53	508	45.60
38	24	2.15	532	47.76
39	13	1.17	545	48.92
40	14	1.26	559	50.18
41	16	1.44	575	51.62
42	24	2.15	599	53.77
43	11	0.99	610	54.76
44	17	1.53	627	56.28
45	9	0.81	636	57.09
46	18	1.62	654	58.71
47	11	0.99	665	59.69
48	15	1.35	680	61.04
49	19	1.71	699	62.75
50	15	1.35	714	64.09
51	12	1.08	726	65.17
52	17	1.53	743	66.70
53	14	1.26	757	67.95
54	14	1.26	771	69.21
55	10	0.90	781	70.11
56	26	2.33	807	72.44
57	15	1.35	822	73.79
58	13	1.17	835	74.96
59	11	0.99	846	75.94
60	13	1.17	859	77.11
61	10	0.90	869	78.01

(*Continued*)

TABLE 2. Continued.

TFI global score	All patients Frequency	Percent	Cumulative frequency	Cumulative percent
62	14	1.26	883	79.26
63	16	1.44	899	80.70
64	9	0.81	908	81.51
65	12	1.08	920	82.59
66	21	1.89	941	84.47
67	6	0.54	947	85.01
68	7	0.63	954	85.64
69	6	0.54	960	86.18
70	11	0.99	971	87.16
71	8	0.72	979	87.88
72	10	0.90	989	88.78
73	5	0.45	994	89.23
74	7	0.63	1,001	89.86
75	6	0.54	1,007	90.39
76	10	0.90	1,017	91.29
77	5	0.45	1,022	91.74
78	8	0.72	1,030	92.46
79	8	0.72	1,038	93.18
80	8	0.72	1,046	93.90
81	4	0.36	1,050	94.25
82	11	0.99	1,061	95.24
83	3	0.27	1,064	95.51
84	7	0.63	1,071	96.14
85	4	0.36	1,075	96.50
86	3	0.27	1,078	96.77
87	1	0.09	1,079	96.86
88	5	0.45	1,084	97.31
89	0	0.00	1,084	97.31
90	3	0.27	1,087	97.58
91	4	0.36	1,091	97.94
92	4	0.36	1,095	98.29
93	1	0.09	1,096	98.38
94	3	0.27	1,099	98.65
95	0	0.00	1,099	98.65
96	3	0.27	1,102	98.92
97	1	0.09	1,103	99.01
98	2	0.18	1,105	99.19
99	3	0.27	1,108	99.46
100	6	0.54	1,114	100.00
Total	1,114	100		

TFI, Tinnitus Functional Index.

results of patients with mild, moderate, severe, and very severe hearing loss were similar and are shown in Figure 3.

In brief, reference values for tinnitus severity have been created for the general population of patients with tinnitus, and the values may be applied to patients regardless of age, gender, hearing status, or tinnitus duration as the effects of these variables are insignificant or very small.

DISCUSSION

Due to its subjective nature, measuring the severity of tinnitus remains a clinical challenge. However, because severity is a vital element of diagnosis and is needed for evaluating the results of intervention, some researchers have attempted to create a scoring system for tinnitus severity. Klockhoff and Lindblom (1967) distinguished three grades of tinnitus taking into account its audibility and disturbance to sleep. A similar

TABLE 3. Variables potentially associated with the TFI global score; results of a linear regression model (N = 1114)

	Unstandardized coefficient	Standard error	Beta standardized coefficient	*p*	Zero-order correlation	Partial correlation	Part correlation
Gender	4.275	1.388	0.091	0.002	0.090	0.092	0.091
Age	0.278	0.055	0.155	<0.001	0.152	0.150	0.149
Tinnitus duration	–0.003	0.008	–0.011	0.715	0.023	–0.011	–0.011
Hearing loss	–0.073	1.701	–0.001	0.966	0.034	–0.001	–0.001

TFI, Tinnitus Functional Index.

categorization was proposed by Baguley et al. (1992). These proposals, although valuable and useful, contain two major drawbacks: their rating depends mainly on the quality of sleep and they are based on the results of a medical and audiological examination. McCombe et al. (2001) considered that the most effective way of measuring tinnitus severity is to use a self-report, psychometrically robust tinnitus-related questionnaire, which seems a more satisfactory approach. However, an individual score has little meaning by itself, and it needs an appropriate context for interpretation. The aim of the present study was to provide a frame of reference by which an individual score could be interpreted relative to others.

As said earlier, Meikle et al. (2012) proposed preliminary limits for interpreting TFI scores: mild (score <25 points), a significant problem (25 to 50), and severe (>50 points). However, such limits seem too coarse, because it would be helpful to distinguish those patients with extremely severe tinnitus. Our findings are that a score near 42 points (M = 41.8) is close to the median (Me = 40.4), and because dispersion is quite high (SD is over 50% of the mean), a criterion level of 50 points for severe tinnitus seems too low. The percentile table shows that 36% of patients achieved a score of more than 50 points, making this a rather large group. Our opinion is that the group with severe tinnitus should be graded more cautiously and we propose a more stringent criterion for high tinnitus severity. The approach of setting categorical boundaries to interpret tinnitus severity seems more sensitive, because it limits the severest rating to a smaller proportion of scores. We suggest that the cutoff for diagnosing high tinnitus severity should be set at 65 points, not at 50 points. Meikle et al. (2012) showed that patients having a big problem with tinnitus scored on average M = 64.0, and patients with a very big problem scored on average M = 77.6. Our findings are similar, with the minimum value for high tinnitus severity being 65 points. Reference values can help with evaluating tinnitus severity and can be used as benchmarks in clinical practice, for example, referral to a specialty tinnitus clinic. Recommendations along these lines have been put forward by the TFI authors (Meikle et al. 2012). Other specialists (Tunkel et al. 2014) also recommend distinguishing patients with bothersome tinnitus from those with nonbothersome tinnitus, giving them different clinical interventions. But it should be kept in mind that tinnitus evaluation is multifaceted, and includes other patient reports, medical examinations, and co-occurring conditions, so TFI score alone will not be an adequate measure of tinnitus severity and should be given alongside other patient data.

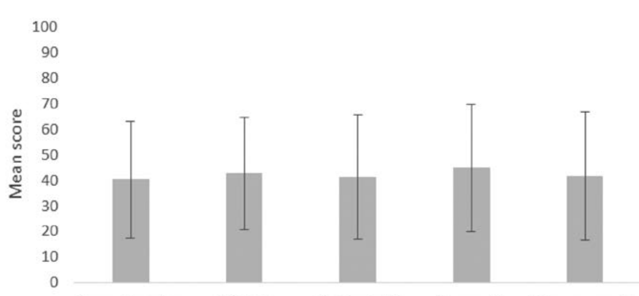

Fig. 3. Mean tinnitus severity measured with TFI global score in patients with normal hearing and hearing loss. Error bars represent SDs. TFI, Tinnitus Functional Index.

The proposed reference values have been established on the basis of the M and SD of the TFI global score. A similar approach was used for the Tinnitus Handicap Inventory (Skarżyński et al. 2020). We recognize that the distribution of the TFI global score is slightly positively skewed and somewhat flattened, so reference values could be slightly affected by skewness and kurtosis. Nevertheless, our results (M and SD) are compatible with those obtained by other researchers who have validated TFI in different settings (Fackrell et al. 2016; Wrzosek et al. 2016; Peter et al. 2017). It can be intuitively appreciated that tinnitus severity might not, in general, be normally distributed because the majority of people with tinnitus are not strongly bothered by it.

Our findings suggest that tinnitus has a significantly greater impact on women than on men, corroborating results of some previous studies (Erlandsson & Holgers 2001; Welch & Dawes 2008); however, data about the role of gender in tinnitus severity are still inconclusive (Pinto et al. 2010). Although we did find a significant difference between men and women, the effect was very small. In fact, we consider it too small to be practically relevant, so we have not established separate reference values for men and women.

The same is true of age. Our study found a significant, albeit small, effect of age on tinnitus severity. This corresponds to Hiller and Goebel's study (2006) which reported that tinnitus loudness and annoyance were perceived as greater in older people (over 50 years old) than in other groups of patients. It is also known that depressive symptoms are more frequent in elderly people (Kok & Reynolds 2017), as well as sleep difficulties associated with tinnitus (Hébert & Carrier 2007). Elderly people may be more sensitive to tinnitus distress and rate self-perceived tinnitus severity as higher than do younger people. But we do not think the small difference between different age groups shown in our study would be considered important by practitioners or researchers, and would not lead to different patient management. We conclude that the small effect of age is clinically insignificant and so we did not seek to establish separate reference values for different age groups.

We found no statistically significant effect of tinnitus duration on tinnitus severity and this result is in line with other studies validating TFI in different languages (Rabau et al. 2014;

Peter et al. 2017). It does not seem to matter how long tinnitus lasts; more important are other factors such as tinnitus loudness and the individual's capacity to cope with the problem.

Tinnitus may occur as an isolated complaint, but it often coexists with hearing loss. Based on a patient's history and the shape of the audiogram, it can be concluded that some cases of hearing loss are caused by exposure to noise. In other patients, genetic causes of hearing loss might be suspected, in which case diagnostic measures such as assessing the occurrence of GJB2 and GJB6 mutations can be undertaken. Another large group of patients are people with idiopathic hearing loss occurring as a consequence of sudden deafness. According to our observations, many patients with tinnitus have the following comorbidities: hypertension, diabetes, thyroid dysfunction (more often as hypothyroidism than hyperthyroidism), degenerative changes in the spine, and disorders of lipid metabolism. If a patient with tinnitus has concomitant hearing loss, then, as a rule, it turns out that hearing loss is likely to be the main problem preventing effective communication, although the patient is not always aware of this. To distinguish whether hearing impairment or tinnitus is the main problem, the THS questionnaire can be used (Henry et al. 2015; Raj-Koziak et al. 2017).

Our results do not show that hearing loss has an effect on tinnitus severity. These findings are in line with some earlier studies (Newman et al. 1996; Noroozian et al. 2017), but are in contrast with the results of some others (Savastano 2008; Oron et al. 2018). We did find that patients with normal hearing had slightly lower tinnitus severity than did patients with hearing loss, but the difference was small, only about two points. Raj-Koziak et al. (2019) considered that hearing loss and tinnitus could overlap in some patients, and these are usually the ones who over-report tinnitus severity. Our results from a large group of patients did not confirm these statements. We evaluated tinnitus severity with a multi-item, self-report questionnaire covering many aspects of tinnitus perception, and it is possible that the association between tinnitus severity and hearing impairment might depend on the particular instrument with which tinnitus is measured. More studies are needed to confirm this supposition and to remove any uncertainty about the relationship between perceived tinnitus and hearing loss.

The major strength of the current study is the large population-based study design and the strong statistical evidence supporting the proposed reference values. Despite these advantages, our study also has some limitations. One of them is that only subjects who attended a tinnitus clinic were involved, so there is the risk of a selection bias with possible underrepresentation of those without a major complaint of tinnitus-related distress. Nonetheless, our patients are referred to our center from all over the country, so they can be considered representative of patients with tinnitus.

We did try to minimize other sources of bias such as nonresponse bias, measurement bias, and analysis bias: to do so we employed rigorous eligibility criteria, used a reliable and valid tool (the TFI), performed thorough analyses, and reported them fully. Another limitation of our study was that although our study group was heterogeneous in terms of gender, age, duration of tinnitus, and hearing loss, our subjects were restricted to being Polish-speaking. There is therefore a need for further studies aimed at establishing more general reference values by gathering data from numerous clinics and different reference populations having broader sociocultural characteristics.

Our findings are similar to those of Meikle et al. (2012), with the difference relating to the names of the categories rather than to the cutoff scores themselves. The current study reinforces the value and robust psychometric qualities of the TFI. In terms of future research, it would be worth tracking some of the patients reported here and calculate (and compare to the original) the TFIs responsiveness to any intervention effects.

To conclude, our study provides reliable reference values that can assist in interpreting scores from the TFI. The findings support the use of the TFI, making it a more useful tool in both clinical and research settings. Setting empirically established benchmarks for levels of tinnitus severity ensures clinicians will have more information with which to make intervention decisions.

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The authors have no conflicts of interest to disclose.

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